Tema 12: Conducción del potencial de acción. Tipos de fibras nerviosas.

Facultad de Biología. Grado de Bioquímica.

Asignatura de “Biofísica”. Dpto. de Fisiología Médica y Biofísica.

Prof.: Rafael J. Montoro

Tema 12. Conducción del potencal de acción. Tipos de fibras nerviosas.

Conducción del potencial de acción
- concepto
- fibras amielínicas
  - cono axónico
  - circuitos locales de corriente

- fibras mielínicas
  - características estructurales
  - circuitos internodales de corriente
    - capacidad eléctrica del axón
    - diámetro
    - distancia internodal

Potencial de acción compuesto

EL POTENCIAL DE ACCIÓN COMO UNIDAD DE INFORMACIÓN

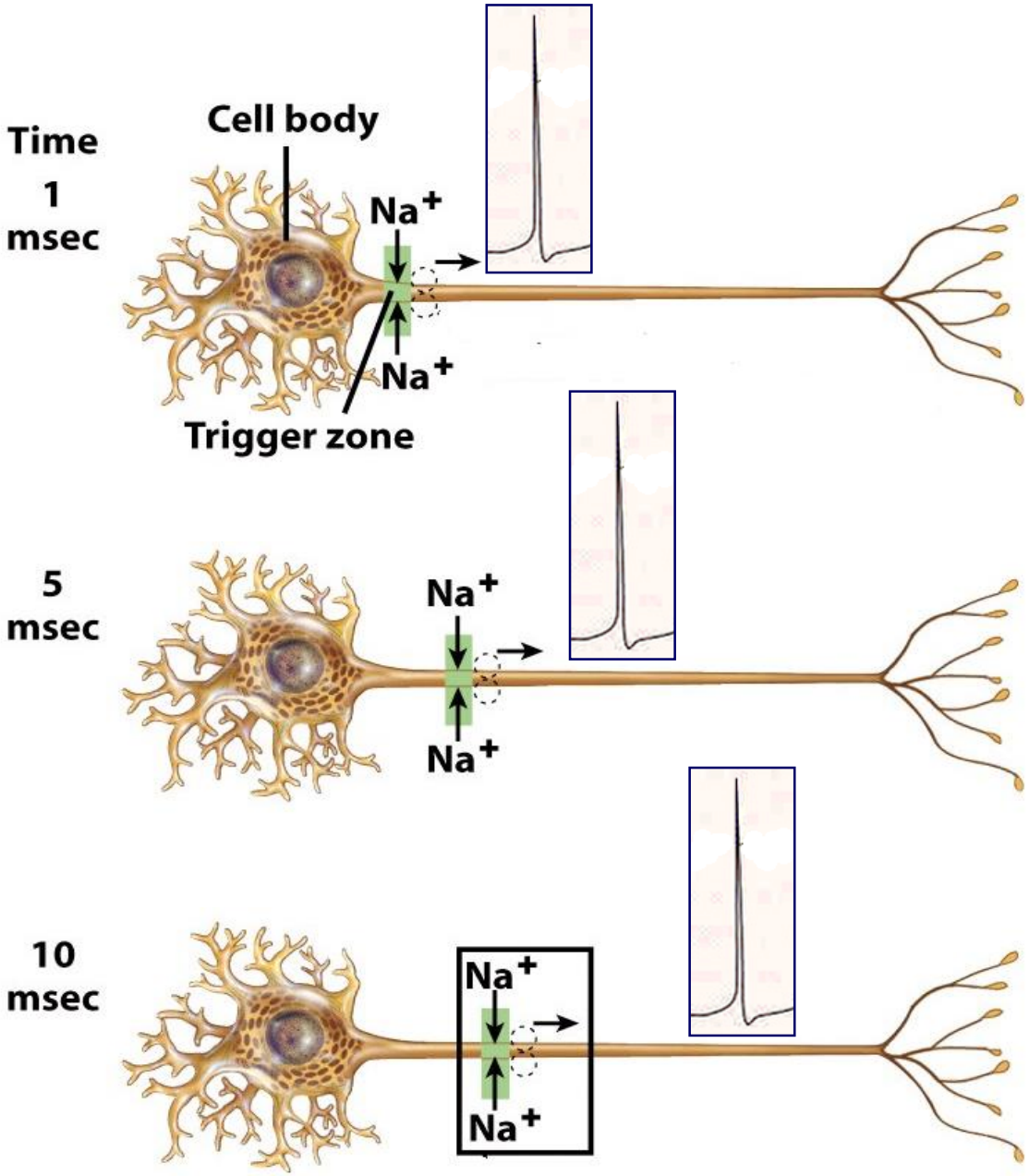
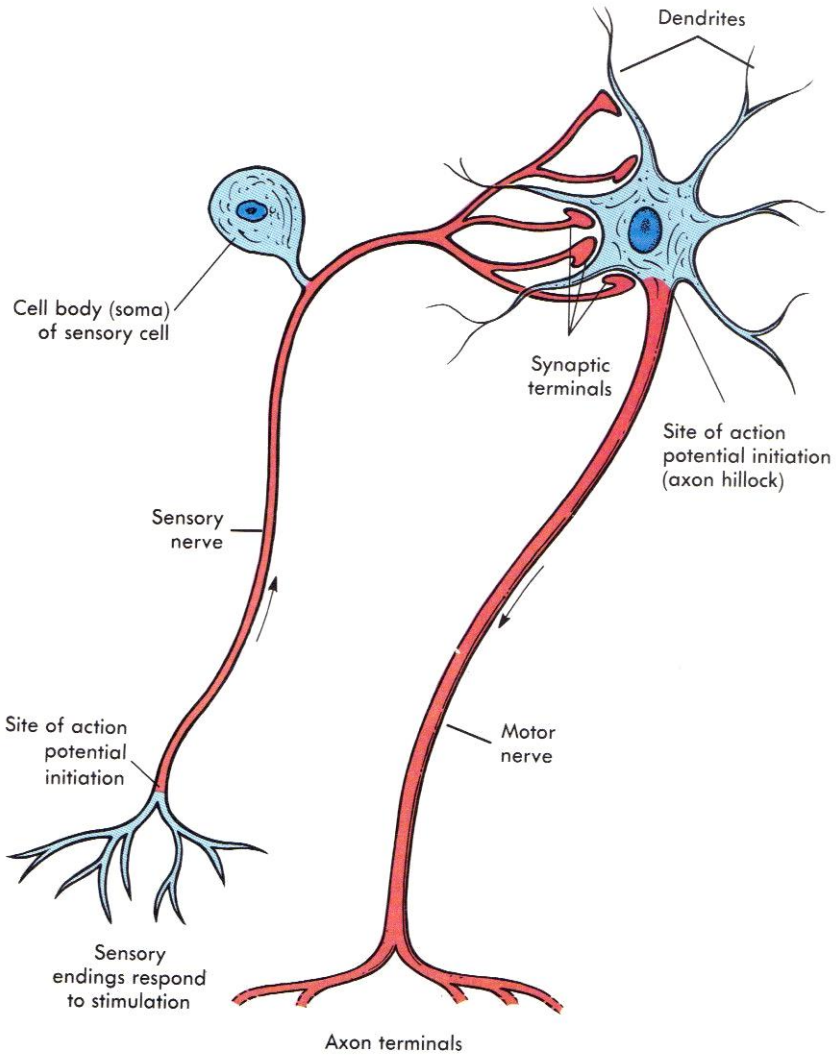


Figure 12-16 Principles of Anatomy
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CONDUCCIÓN DEL POTENCIAL DE ACCIÓN



CONO AXÓNICO

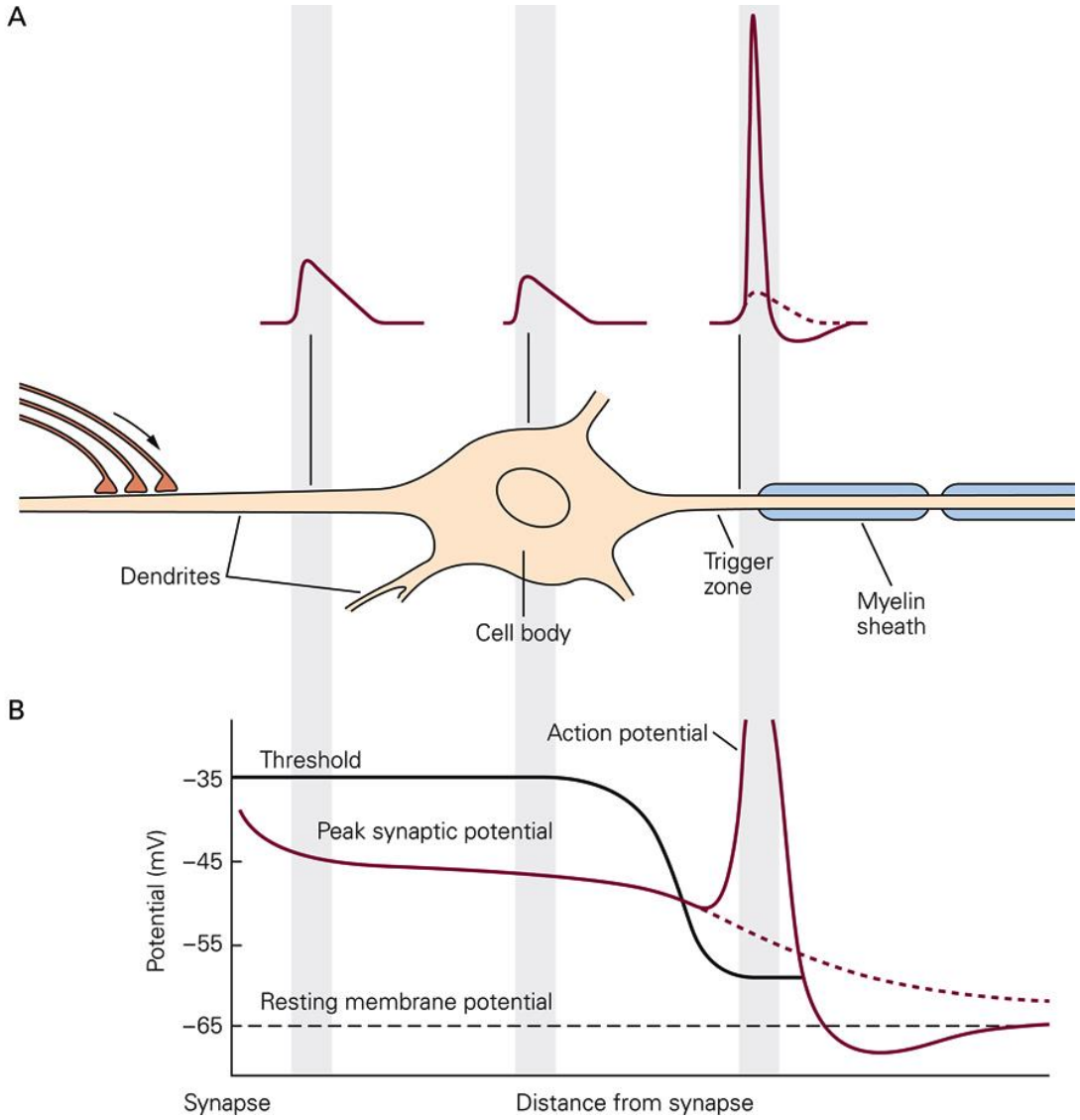


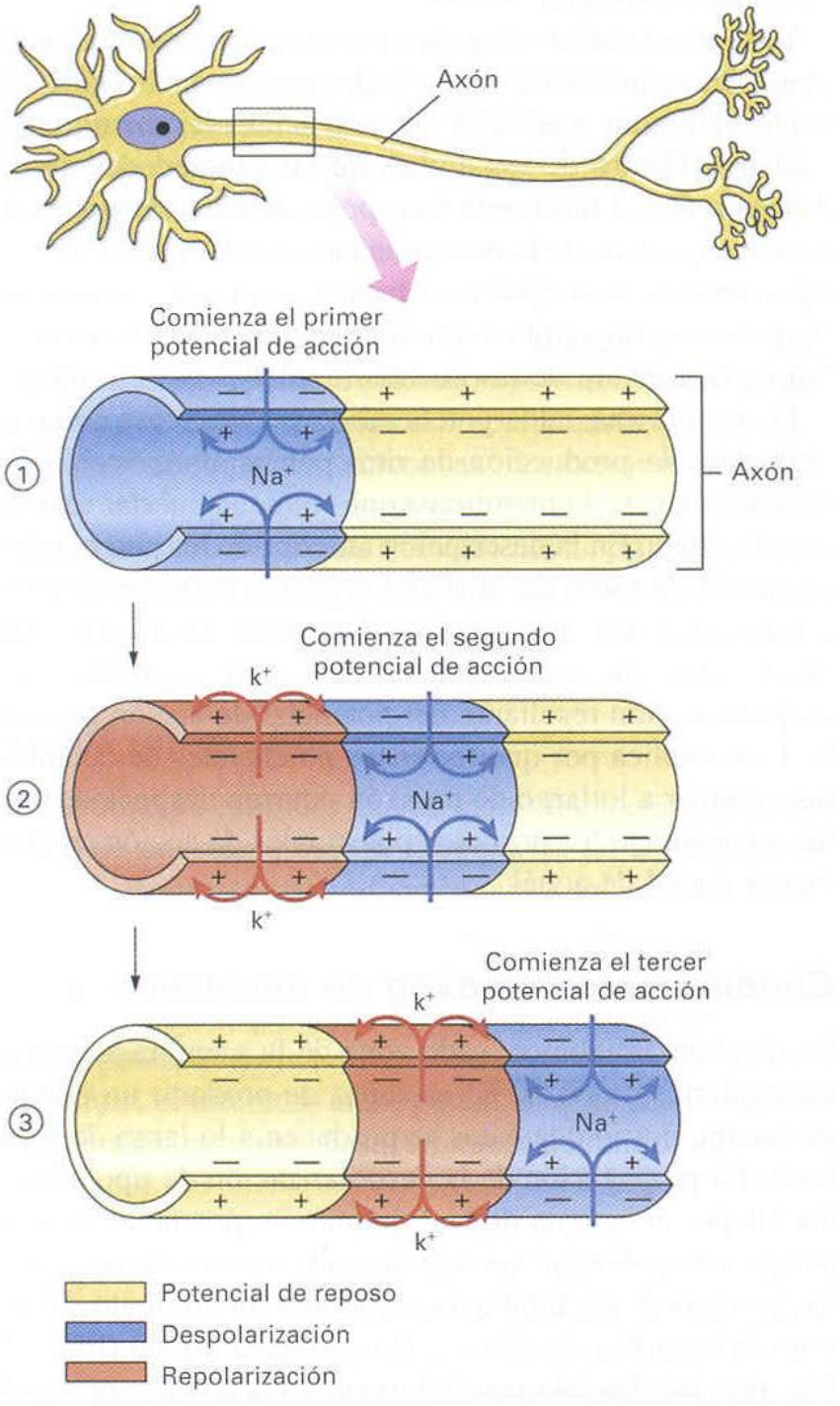
Figure 10–13 A synaptic potential arising in a dendrite can generate an action potential at the axon initial segment. (Adapted, with permission, from Eckert et al. 1988.)

A. An excitatory synaptic potential originating in the dendrites decreases with distance as it propagates passively to the soma. Nevertheless, an action potential can be initiated at the trigger zone (the axon initial segment) because the density of the Na^+ channels in this region is high, and thus the threshold is low.

B. Comparison of the threshold for initiation of the action potential at different sites in the neuron (corresponding to drawing A). An action potential is generated when the amplitude of the synaptic potential exceeds the threshold. The **dashed line** shows the decay of the synaptic potential if no action potential is generated at the axon initial segment.

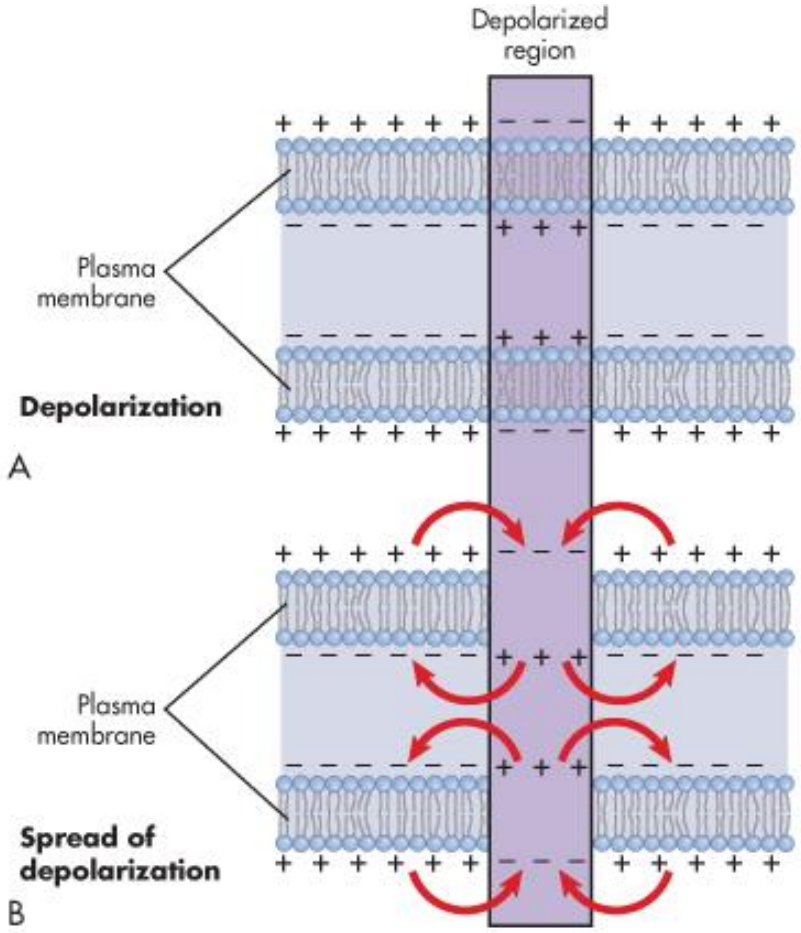
Tomado de Principles of Neural Science, Kandel y cols.,McGraw-Hill, 2013.

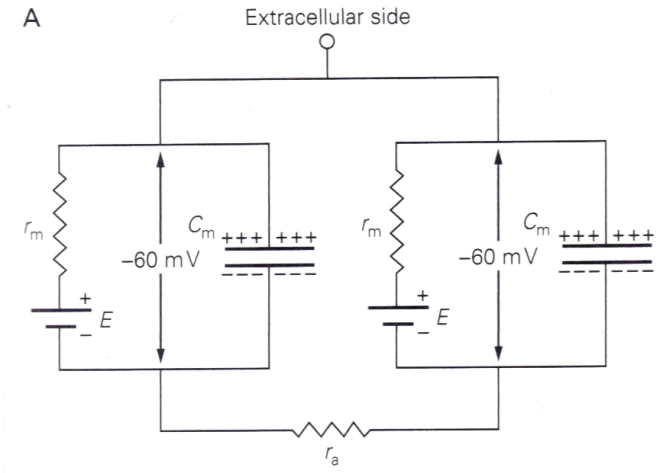
PROPAGACIÓN DEL POTENCIAL DE ACCIÓN EN FIBRAS AMIELÍNICAS

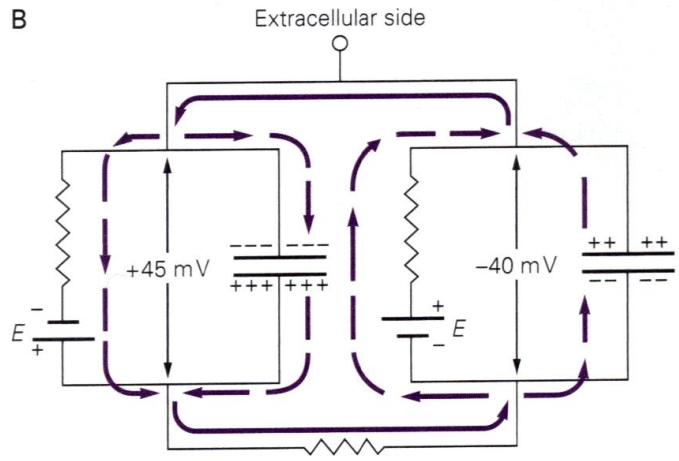


tomado de Fisiología Humana, Fox 7ed

CIRCUITOS LOCALES DE CORRIENTE



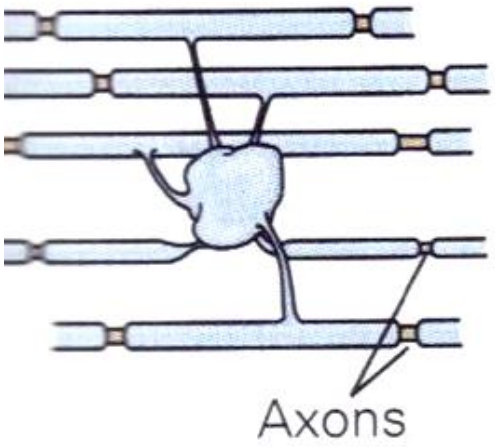




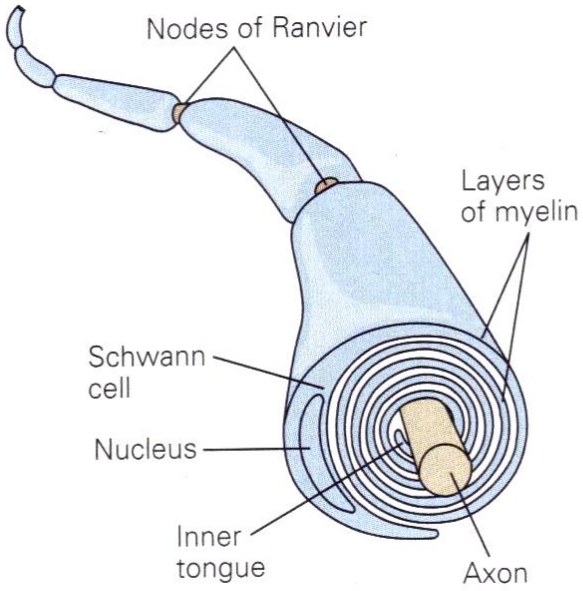
circuitos locales de corriente entre segmentos contiguos.

CÉLULAS GLIALES QUE FORMAN LA VAINA DE MIELINA

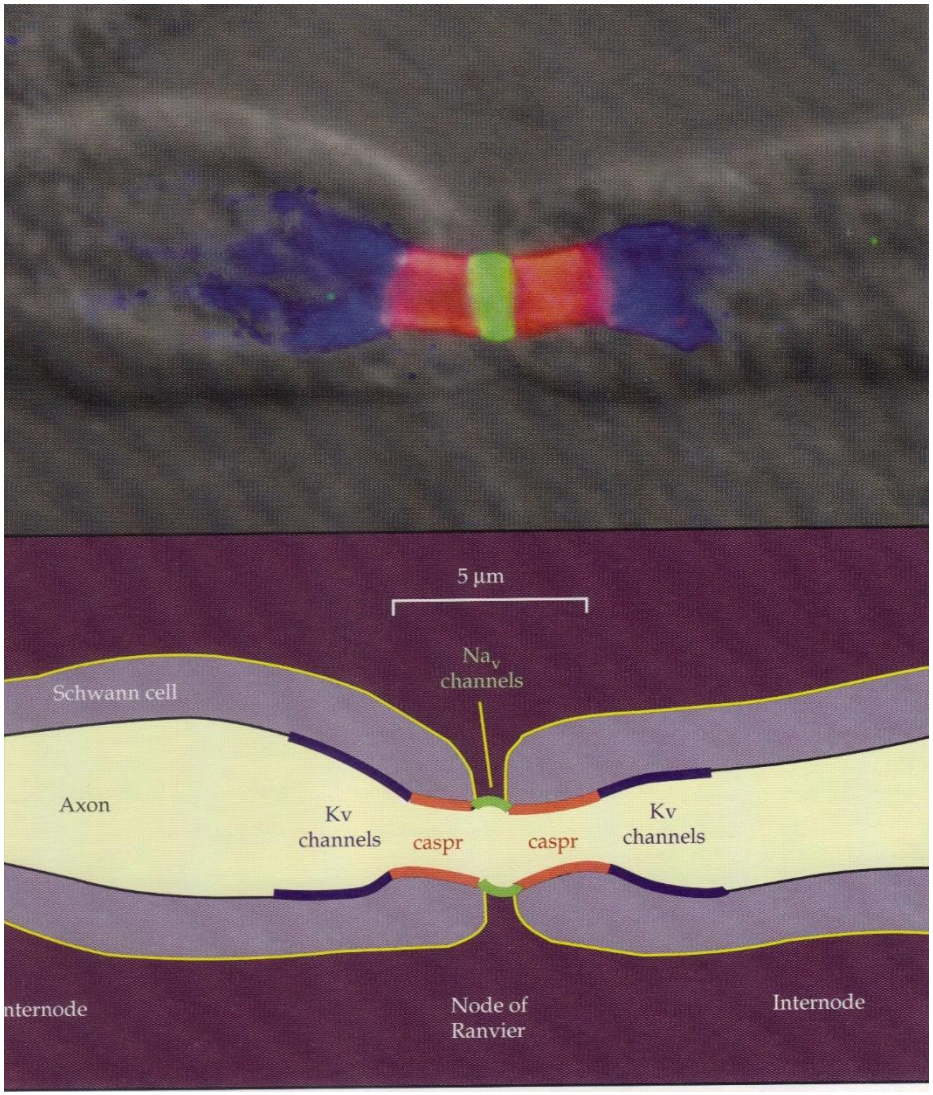
Oligodendrocitos
SNC



Células de Schwann
SNP

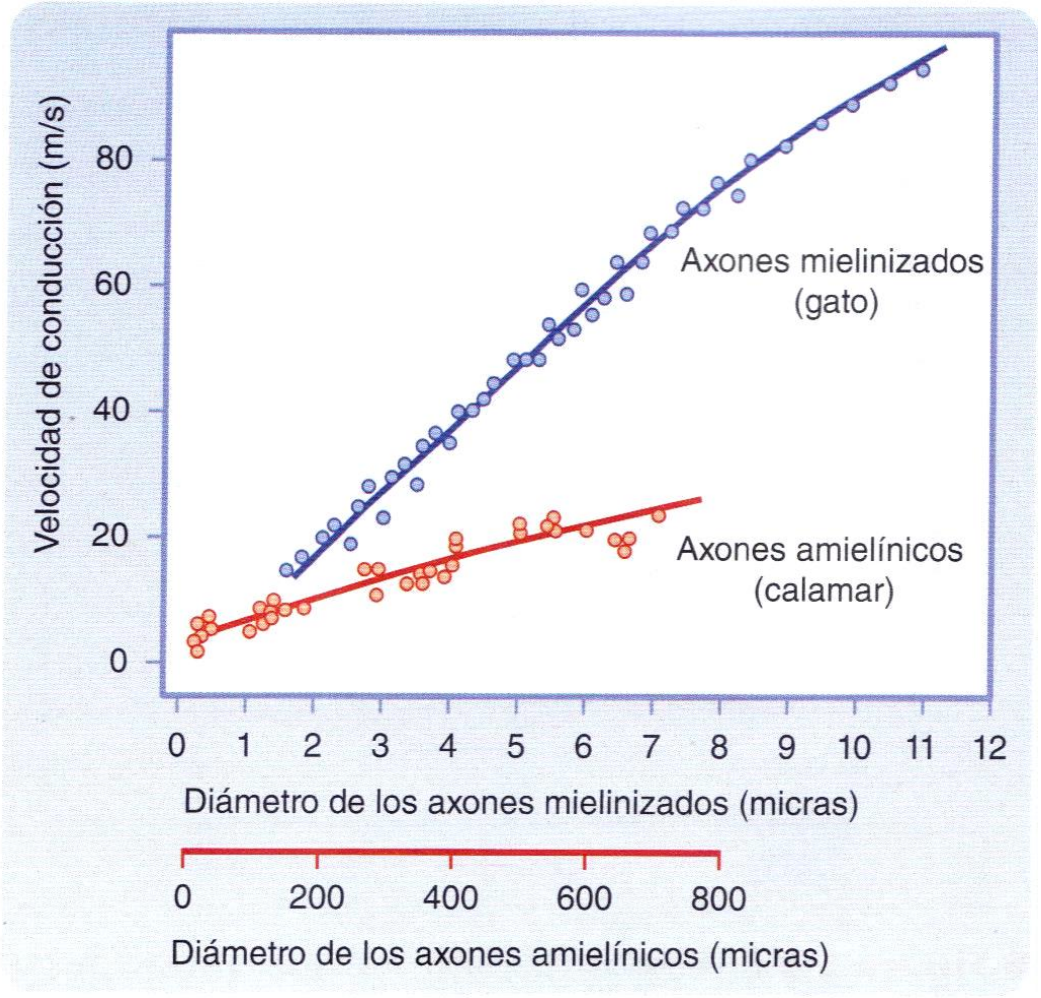


NODO DE RANVIER



Tomada de “Ion channels of excitable membranes”. B. Hille. 3rd ed. Plate 1.

VELOCIDAD DE CONDUCCIÓN DEL POTENCIAL DE ACCIÓN

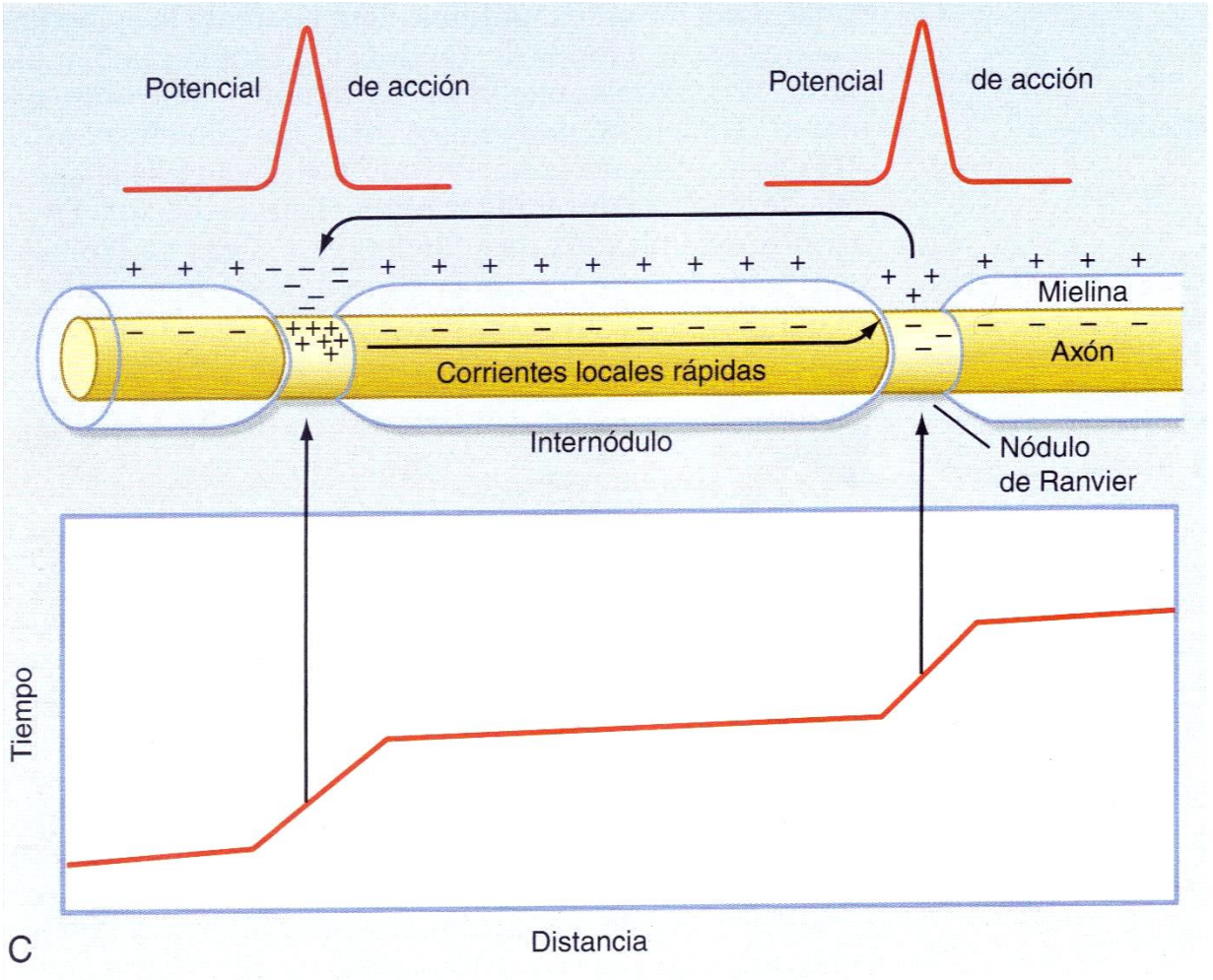


$$\lambda = \sqrt{\frac{r_m}{r_a}}$$

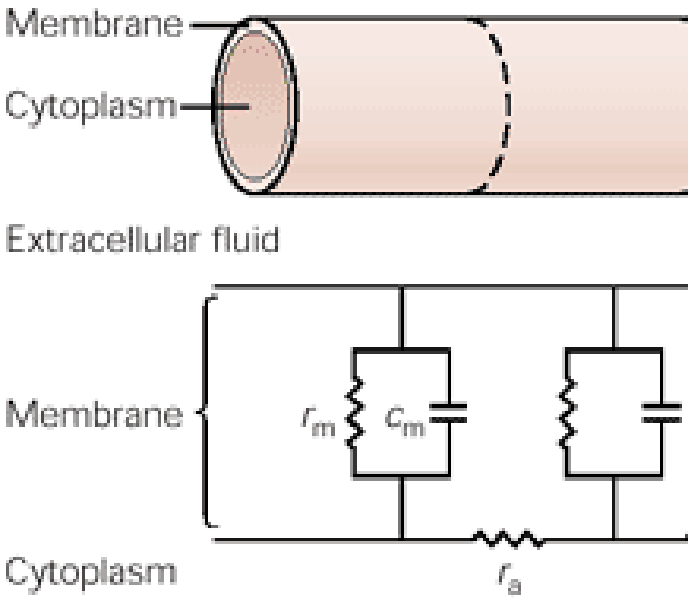
Tomada de Berne et al. Fisiología. 6ª ed.

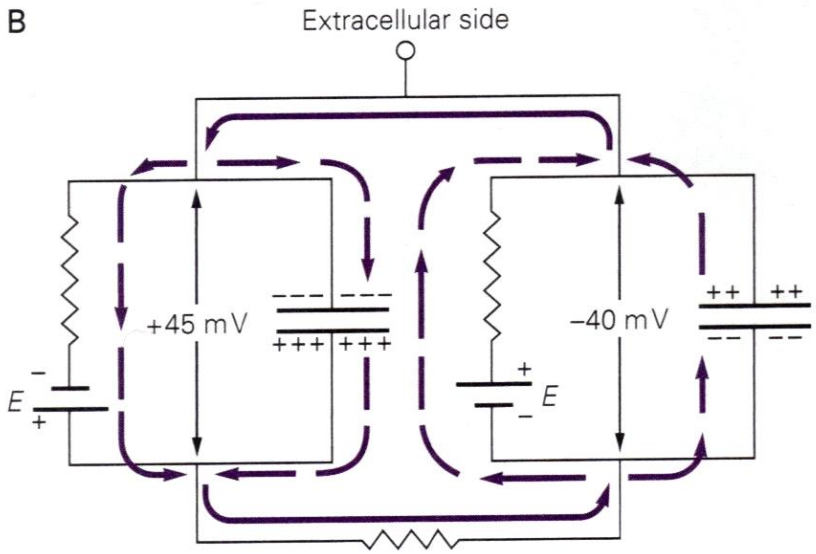
CIRCUITOS INTERNODALES DE CORRIENTE

$$\lambda = \sqrt{\frac{r_m}{r_a}}$$



EFECTO DE LA DISMINUCIÓN DE LA CAPACIDAD EN FIBRAS MIELÍNICAS





$$c_m = \varepsilon \frac{S}{d}$$

$$\Delta V = \Delta Q / C$$

CONSECUENCIAS DE LA DESMIELINIZACIÓN DE FIBRAS

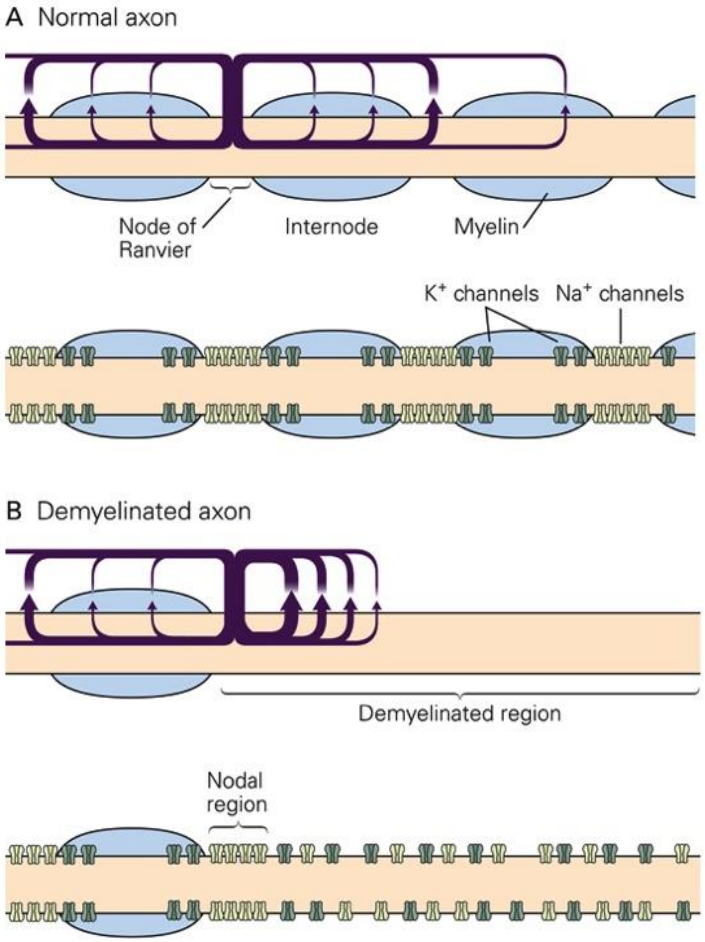


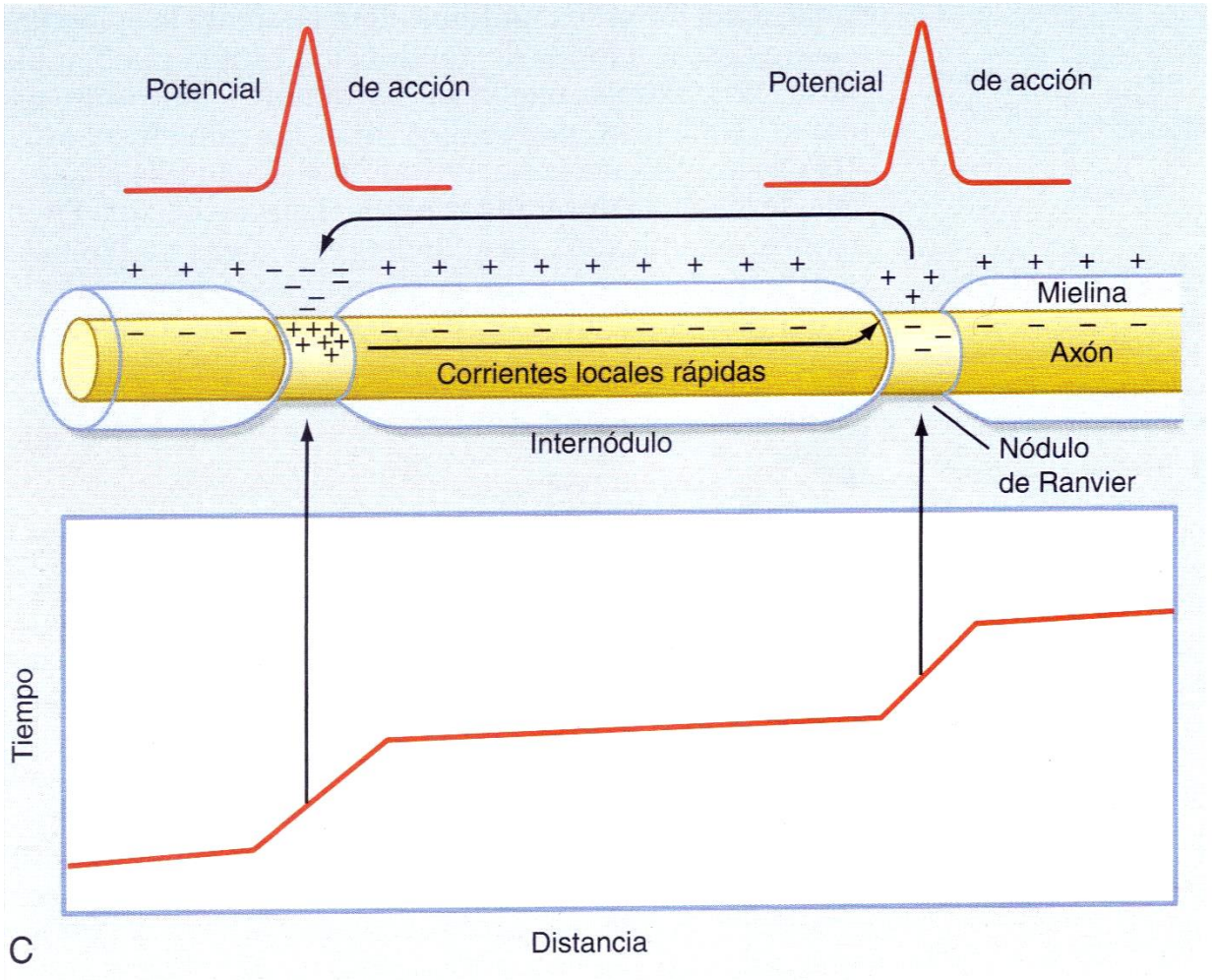
Figure 6–18 Action potentials in myelinated nerves are regenerated at the nodes of Ranvier.

A. The densities of capacitive and ionic membrane currents (membrane current per unit area of membrane) are much higher at the nodes of Ranvier than in the internodal regions. (In the figure the density of membrane current at any point along the axon is represented by the thickness of the arrows.) Because of the higher capacitance of the axon membrane at the nodes, the action potential slows down as it approaches each node and thus appears to skip rapidly from node to node as it propagates from left to right.

B. In regions of the axon that have lost their myelin, the spread of the action potential is slowed down or blocked. The local-circuit currents must discharge a larger membrane capacitance and, because of the shorter length constant (caused by the low membrane resistance in demyelinated stretches of axon), they do not spread well down the axon.

DISTANCIA INTERNODAL

$$\lambda = \sqrt{\frac{r_m}{r_a}}$$



Tomada de Principles of Neural Science. Kandel et al. McGraw Hill. 5ª ed. Fig. 6-18.

TIPOS DE FIBRAS

Table 22-2 Afferent Fiber Groups in Peripheral Nerves

	Muscle nerve*	Cutaneous nerve*	Fiber diameter (μm)	Conduction velocity (m/s)
Myelinated				
Large	I	Aα	12–20	72–120
Medium	II	Aβ	6–12	36–72
Small	III	Aδ	1–6	4–36
Unmyelinated	IV	C	0.2–1.5	0.4–2.0

*Sensory nerves in muscle are classified according to their fiber diameters. Sensory afferents in cutaneous nerves are classified by conduction velocities. The types of receptors innervated by each type of afferent are listed in Table 22-1.

POTENCIAL DE ACCIÓN COMPUESTO

